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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (CE) (Sem.–6)

STRUCTURAL ANALYSIS AND DESIGN

Subject Code :PECE-603D-18

M.Code : 79403

Date of Examination : 19-12-2022

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

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SECTION-A

1. Write briefly :

a) What are indeterminate structures?

b) What are the different types of staircase?

c) Write down different types of retaining walls.

d) Explain Muller Breslau Principle.

e) Differentiate between Portal and Cantilever method.

f) Where combined foundation is used?

g) What is Bracket Connection? Describe in brief.

h) What is Plate Girder? Draw a neat Sketch.

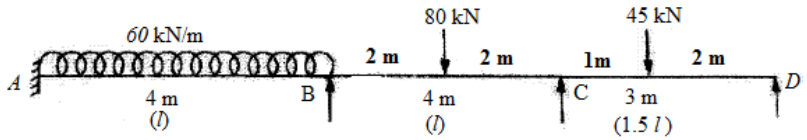
i) Write a short note on type of roof trusses.

j) Differentiate between seated and framed connections.

SECTION-B

2. Draw a neat sketch of the isolated foundation. Also write down the steps to design it.

3. Find the support moments at A, B, C, D for the continuous beam shown in Figure.



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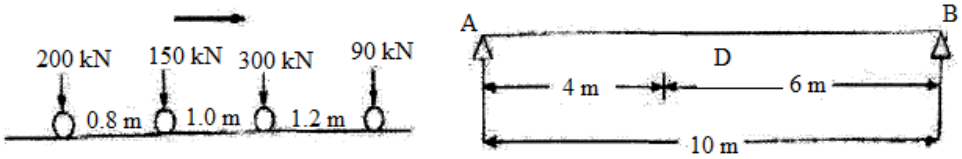
4. The wheel loads shown in figures a) roll over along a beam of span 10 meters. Find the maximum bending moment which can occur at a section 4 meters from the left end



5. Design a single stair to reach a roof slab at the height of 2.7m. Rise and tread of the steps may be taken as 180 mm and 250 mm respectively. The stairs shall be lm wide*. Use M20 concrete and Fe 415 steel.

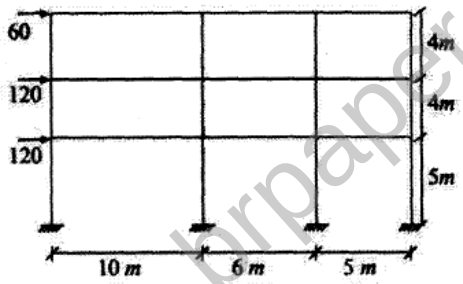
6. A non Sway column in a building frame with flexible joints is 4m high and subjected to the following load and moment:

   Factored axial load 500kN, Factored moment at top of Column = 27kNm, Factored moment at bottom of column = 45kNm. Design a suitable beam column assuming Fe 250. Take the effective length of column as 0.8 L along both the axes.

SECTION-C

7. Design s roof truss, rafter tracing and purlin for an industrial building located at Guwahati with a span of 20m and length of 50m. The roofing is galvanized iron sheeting. Basic wind speed is 50m/s. Building is class B building with a clear height of 8m at the eaves.

8. Assuming area of exterior columns to be half of Interior columns. Analyse the frame using Cantilever method.



9. Design a reinforced concrete combined rectangular footing for two columns each 400 mm × 400mm located 4m between centres and each column carries an axial loads of 1200 kN. The projection of footing parallel to the length of footing beyond the axis of each column is lm. The SBC of soil is 275kN/m^2. Use M20 concrete and Fe415 steel.

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